



Executive Post Graduate Certificate in Applied AI and Machine Learning

Module 1, Lecture 4

Decision Tree, Ensemble Classification with Bagging and Random Forest

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Features for Classification

- We can extract as many features as possible from each datapoint
- Only some features are useful for classification
- We cannot classify cats and dogs based on the number of legs

  - all examples have same value for the feature!

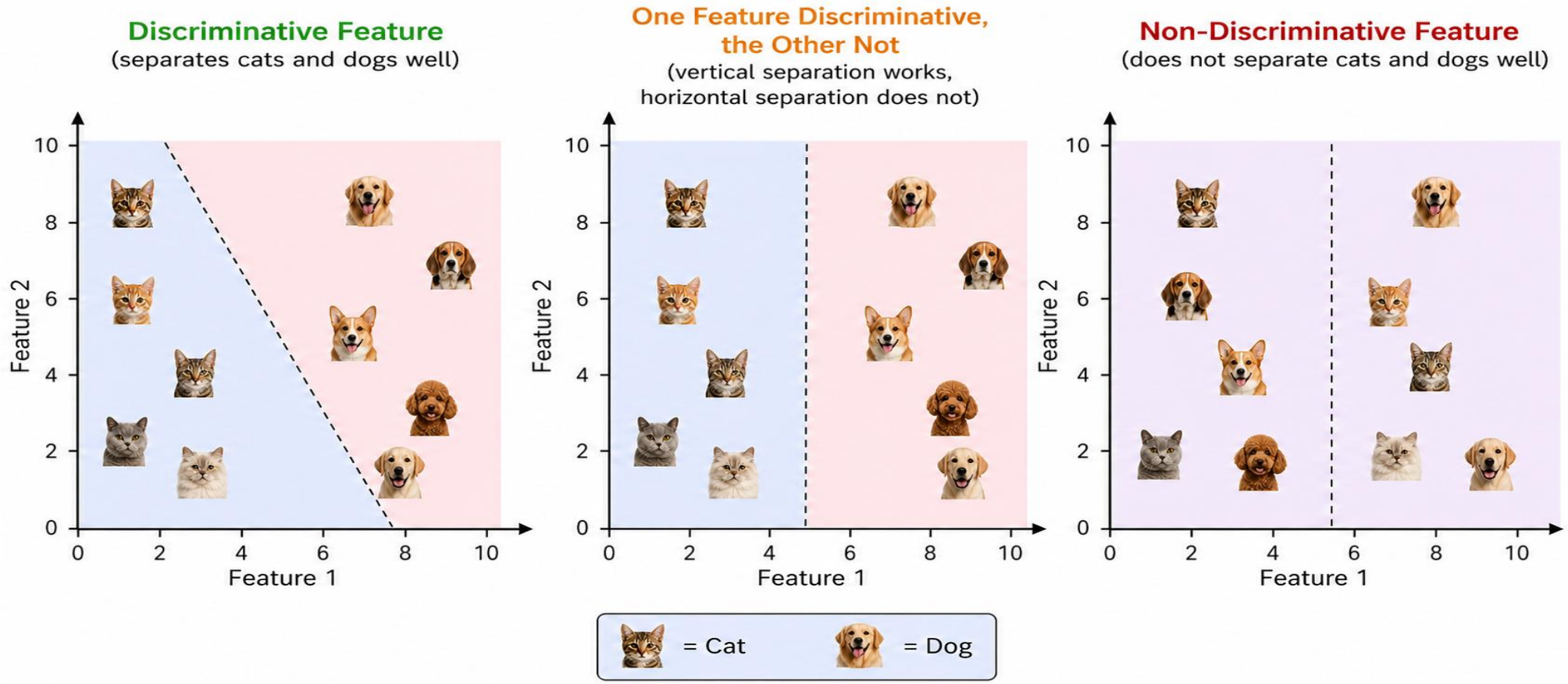
- Or based on colours!

  - significant overlap between characteristic colours of both sets

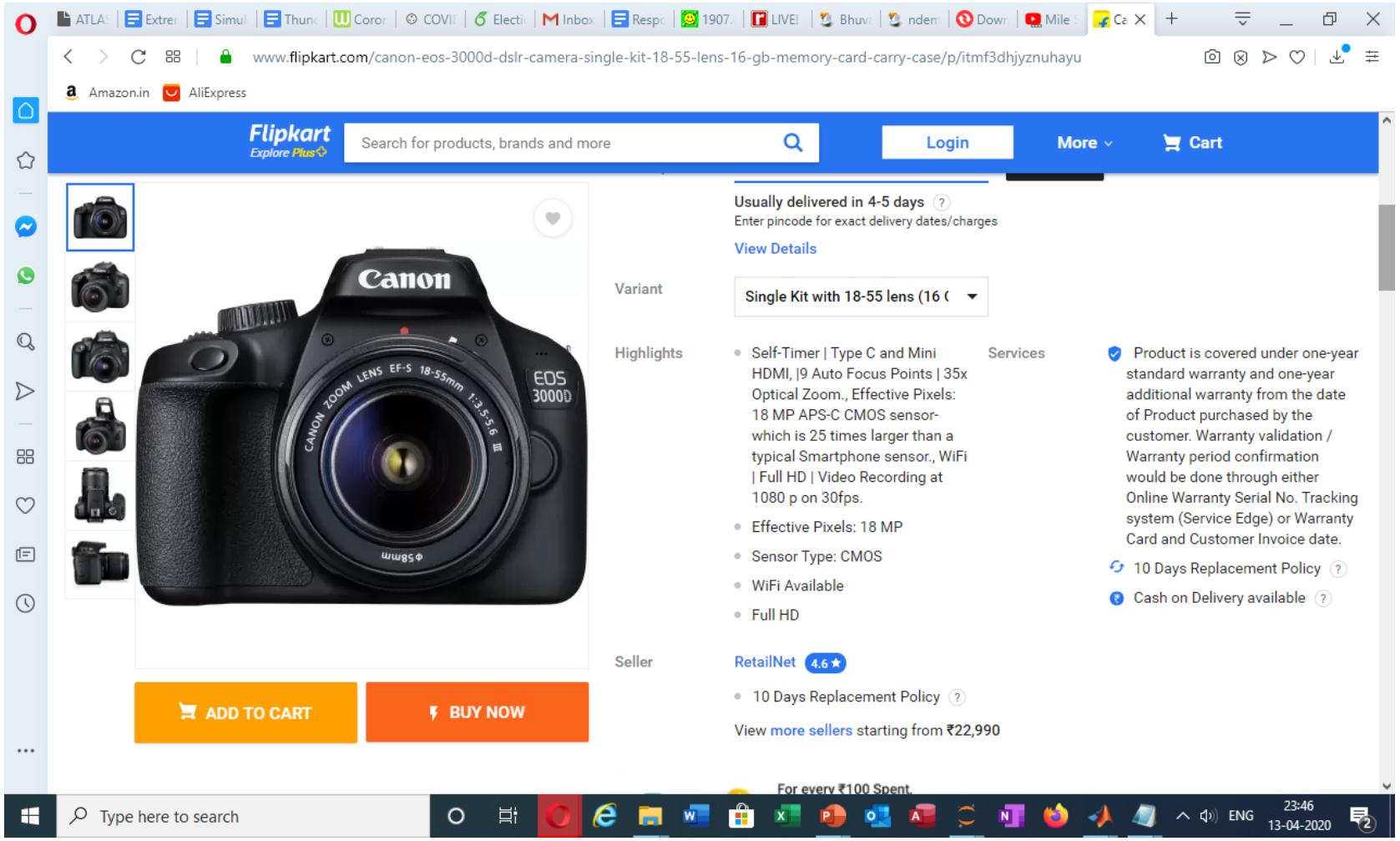
Can we characterize which features are suitable for classification?

Discriminative vs Non-discriminative Features

- House attributes: location, no. of floors, color, building material…..
- Are all of them useful to predict the price of the house?



Predicting Product Ratings



How much rating will a particular user give this camera out of 5?

Probably depends on features!

Which features does the user like?

Source: Flipkart website

amazon.com



Procus Viper 16MP 4K HD Action Camera Waterproof with Wi-Fi

Visit the PROCUS Store

4.3 ★★★★☆ 995 ratings | 240 answered questions

-41% ₹4,699.00

M.R.P.: ~~₹7,999.00~~

Fulfilled

Inclusive of all taxes

EMI starts at ₹221. No Cost EMI available EMI options

Save Extra with 4 offers

Cashback (6): Amazon Pay Reward - Earn 10% cashback upto Rs. 100 on minimum purchase of ₹500 using postpe c... | See All

No Cost EMI: Avail No Cost EMI on select cards for orders above ₹3000 | Details

See 2 more

Pay on Delivery | 7 Days | Amazon | 6 Month

Title:

Presenter:

Date: Time:

Your job classification: ☐ Classified ☐ Professional/Technical ☐ Administrator ☐ Faculty

Please circle the appropriate response for each statement:

	Excellent	Good	Fair	Poor
1. The relevance of this topic to me was	4	3	2	1
2. The usefulness of materials was	4	3	2	1
3. The effectiveness of the presenter was	4	3	2	1
4. I expect the future usefulness of this topic to be	4	3	2	1
5. My overall evaluation of this session is	4	3	2	1

Your Account > Packaging Feedback

Rate Amazon's Packaging

Did the packaging protect your items adequately? ☆☆☆☆☆ Protection — 1 star = Poor; 5 stars = Excellent

Was the box size and packaging appropriate for the items?

- ◯ Too Small
- ◯ About Right
- ◯ Too Big
- ◯ Way Too Big

Rate Item's Packaging

[redacted] ☆☆☆☆☆ Ease of Opening — 1 star = Very Difficult; 5 stars = Very Easy

Central Railway

Annexure E3 (A)

FEEDBACK FORM

"On-Board Housekeeping Services" - Indian Railways

AC COACH S. No:_____

Dear Passenger,

Our endeavor is to provide you the most hygienic On Board Housekeeping Services. Your valuable feedback would help us improve further.

Kindly spare few minutes in rating the areas as given in table below:
Ratings

5 = Excellent, 4 = Very Good, 3 = Good, 2 = Average, 1 = Poor

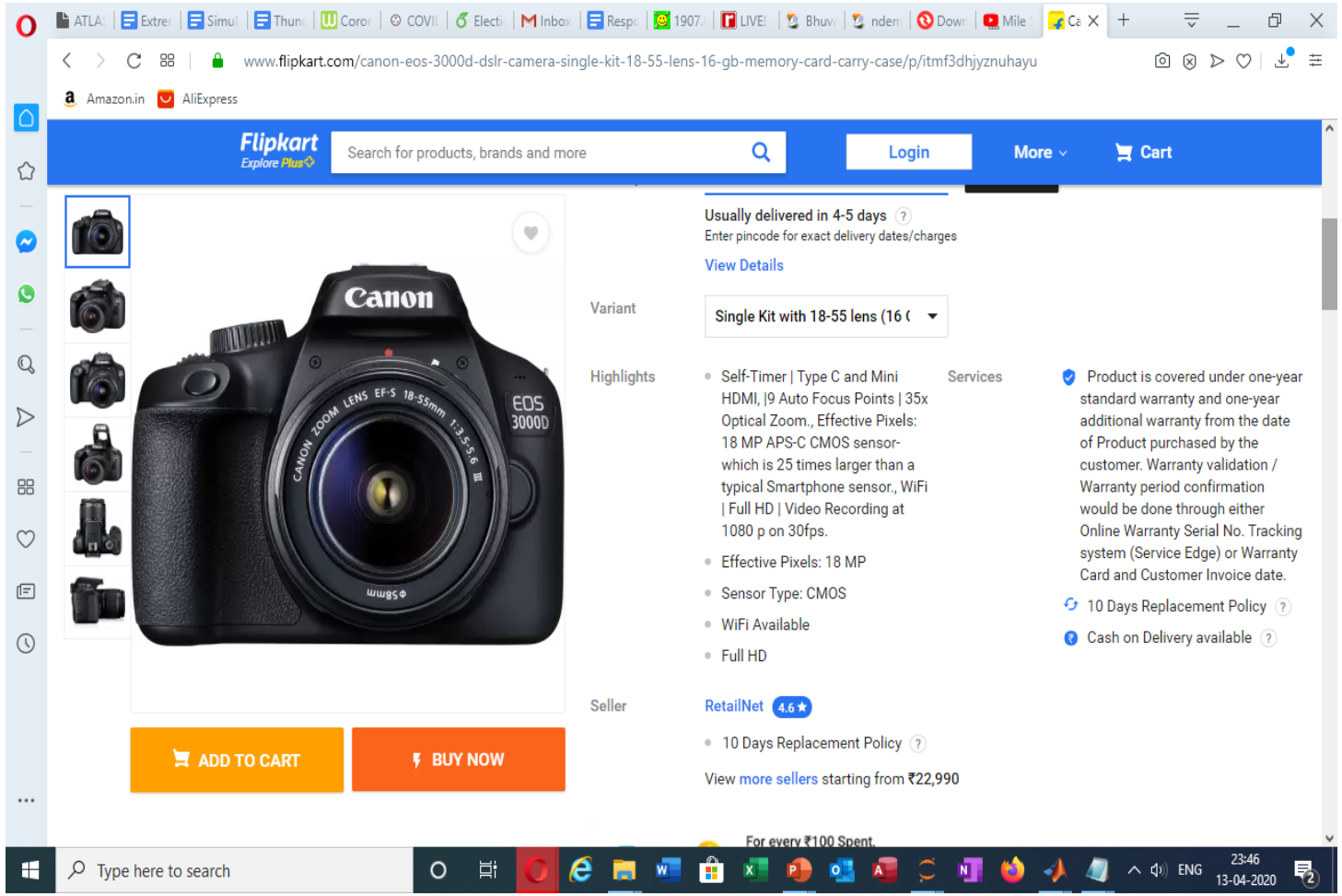
Passenger Feedback - AC Coaches

Sr. No.	Areas of Cleaning / Services	5	4	3	2	1
	Please mark (✓) in space					
1	Cleaning / Washing of Toilet floor and commode pan					
2	Dry Cleaning of Toilet Floor					
3	Cleaning of Mirror, shelf, wall panels and other fittings in Toilets					
4	Cleaning of Wash Basin in Toilets and Doorways					
5	Cleaning of Doorway Area					
6	Cleaning of Vestibule Area including entrance to toilets					
7	Cleaning of Passenger compartments					
8	Cleaning of Passenger aisle area					
9	Cleaning of Window Glasses on Platform side					
10	Cleaning of Dust Bins of coaches					
11	Disinfection and provision of Deodorant in toilets					
12	Spraying of air freshener in compartments					
13	Spraying of Mosquito Repellent					
14	Replenishment of Liquid Soap in Coach toilets					
15	Replenishment of Tissue Paper Roll in Western style Coach toilets					
16	Collection of Garbage and disposal in Poly Bags duly segregate as Biodegradable / Non biodegradable					
17	Behaviour of Janitors / Supervisor					
18	Hygiene & Cleanliness of Janitors / Supervisor including their uniform					
	Scores*					
	Passenger Satisfaction Index (PSI)*					

***Not to be filled by the passenger**

Image source: Google Images

Discrete Product Ratings based on Discrete Features



How much rating will a particular user give this camera out of 5?

Probably depends on features!

Which features does the user like?

Source: Flipkart website

Feature Selection

- The user has exactly 5 options: 1, 2, 3, 4 or 5 stars!
- Her choice depends on the different features of the product!
- But she may consider some features to be more important than others !
- Which features determine her vote?

Feature Selection

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- Which features determine her vote?

Company	Color	Resolution	Video Rate	Price	Her Rating
C1	Black	10 MP	25 fps	$200	2
C1	White	15 MP	25 fps	$250	2
C2	White	12 MP	30 fps	$250	4
C1	Black	15 MP	30 fps	$300	3
C2	Black	20 MP	25 fps	$400	3
C2	White	12 MP	50 fps	$500	5
C2	Black	15 MP	30 fps	$250	????

Feature Selection

- The user has 5 exactly options: 1, 2, 3, 4 or 5 stars!
- Her choice depends on the different features of the product!
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C2	Black	20 MP	25 fps	$400	3
C2	White	12 MP	50 fps	$500	5
C2	Black	15 MP	30 fps	$350	4

Feature Selection

- The user has 5 exactly options: 1, 2, 3, 4 or 5 stars!
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- But she may consider some features to be more important than others !
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C1	Black	15 MP	30 fps	$300	3
C2	Black	20 MP	25 fps	$400	3
C2	White	12 MP	50 fps	$500	5
C2	Black	15 MP	30 fps	$350	4

Decision Tree for Feature Selection

- Which features does she consider as important while rating?
- Let's look at her history of rating 100 cameras!

Which feature is more important for ratings - company or color???

Rating	Count
1	21
2	24
3	18
4	20
5	17

Overall, Count=100

Rating	Count
1	15
2	18
3	10
4	5
5	6

Company = C1, Count=54

Rating	Count
1	6
2	6
3	8
4	15
5	11

Company = C2, Count=46

Rating	Count
1	15
2	20
3	13
4	12
5	10

Color=Black, Count=70

Rating	Count
1	6
2	4
3	5
4	8
5	7

Color=White, Count=30

What's a discriminative feature?

- Company ={C1, C2}, Price = real number, Y = {LOW (1-3), HIGH (4-5)}

	COMPANY=C1	COMPANY=C2	
#(Y=LOW)	43	20	63
#(Y=HIGH)	11	26	37
Total	54	46	100

What's a discriminative feature?

- Company ={C1, C2}, Price = real number, Y = {LOW (1-3), HIGH (4-5)}

	Price<300	Price >=300	
#(Y=LOW)	45	18	63
#(Y=HIGH)	25	12	37
Total	70	30	100

What's a discriminative feature?

- Company ={C1, C2}, Price = real number, Y = {LOW (1-3), HIGH (4-5)}

	Price<500	Price >=500	
#(Y=LOW)	55	8	63
#(Y=HIGH)	35	2	37
Total	90	10	100

What's a discriminative feature?

- Prob(Y = HIGH | COMPANY = C1) = 11/54 ~ 0.2 [Easy to decide]
- Prob(Y = HIGH | COMPANY = C2) = 26/46 ~ 0.55

- Prob(Y = HIGH | PRICE < 300) = 25/70 ~ 0.36
- Prob(Y = HIGH | PRICE >= 300) = 12/30 = 0.4

- Prob(Y = HIGH | PRICE < 500) = 35/90 ~ 0.4
- Prob(Y = HIGH | PRICE >= 500) = 2/10 ~ 0.2 [Easy to decide]

[Very few examples]

What's a discriminative feature?

- Prob(Y = HIGH | COMPANY = C1) = 11/54 ~ 0.2 [Easy to decide]
- Prob(Y = HIGH | COMPANY = C2) = 26/46 ~ 0.55

COMPANY: good feature

- Prob(Y = HIGH | PRICE < 300) = 25/70 ~ 0.36
- Prob(Y = HIGH | PRICE >= 300) = 12/30 = 0.4

PRICE<300: bad feature

- Prob(Y = HIGH | PRICE < 500) = 35/90 ~ 0.4
- Prob(Y = HIGH | PRICE >= 500) = 2/10 ~ 0.2 [Easy to decide]

PRICE<500: doubtful feature [Very few examples]

Decision Tree Algorithm

- Idea: identify the “most discriminative” feature, use it to classify!
- Problem 1: How to quantify “discriminative-ness”?
- Problem 2: What if no feature is very discriminative?

Decision Tree Algorithm

- Idea: identify the “most discriminative” feature, use it to classify!

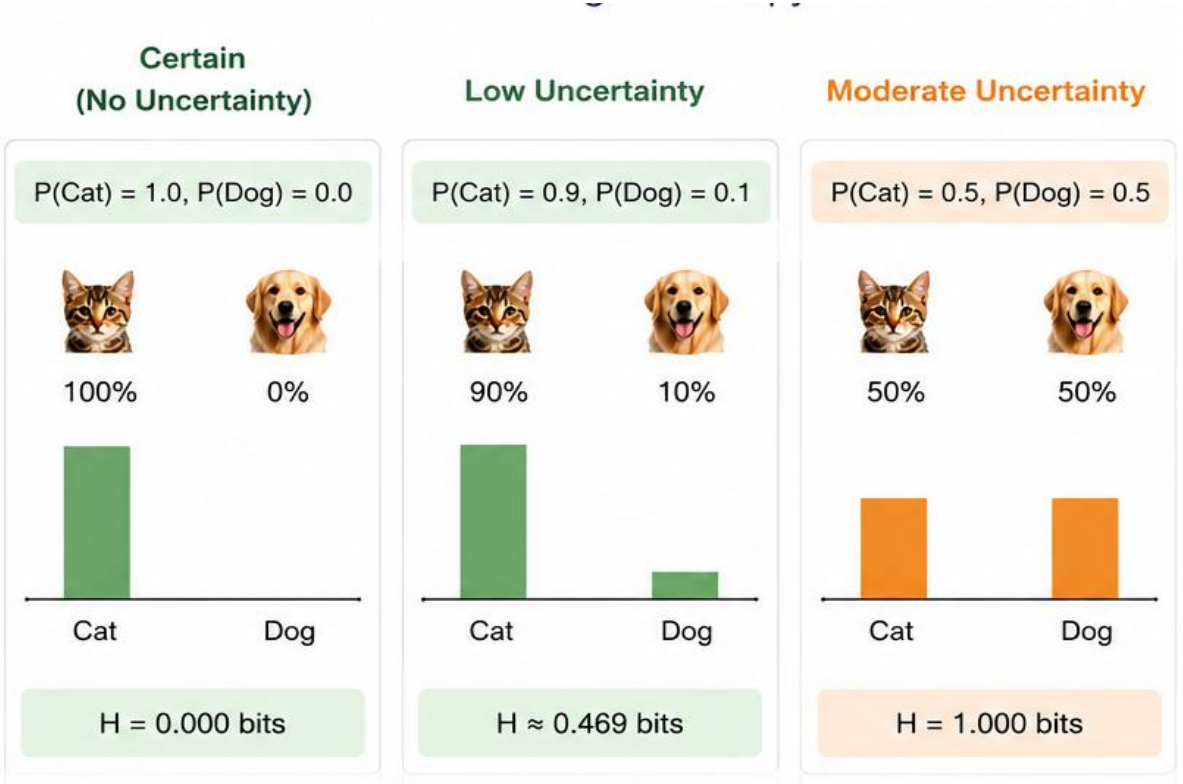
- Problem 1: How to quantify “discriminative-ness”?
  - entropy!
- Problem 2: What if no feature is very discriminative?
  - try a sequence of features!

Entropy: measure of discriminativeness

- P(Y=1) = 0.5, p(Y=2) = 0.5 : low discriminative ability
- P(Y=1) = 0.9, p(Y=2) = 0.1 : high discriminative ability

$$H = -\sum_i p_i(\log_2 p_i)$$

- Case 1: H = 1 (0.69)
- Case 2: H = 0.5 (0.33)



Feature selection based on entropy

	COMPANY=C1	COMPANY=C2	NO SPLIT
#(Y=LOW)	43	20	63
#(Y=HIGH)	11	26	37
Entropy	0.51	0.68	0.66

Information gain =

Original Entropy – (Split1_size*Split1_ entropy + Split2_size*Split2_ entropy)

0.66 – (54/100*0.51 + 46/100*0.68) ~ 0.07

Feature selection based on entropy

	PRICE<300	PRICE>=300	NO SPLIT
#(Y=LOW)	45	18	63
#(Y=HIGH)	25	12	37
Entropy	0.65	0.67	0.66

Information gain =

Original Entropy – (Split1_size*Split1_ entropy + Split2_size*Split2_ entropy)

0.66 – (70/100*0.65 + 30/100*0.67) ~ 0!!

Feature selection based on entropy

	PRICE<500	PRICE>=500	NO SPLIT
#(Y=LOW)	55	8	63
#(Y=HIGH)	35	2	37
Entropy	0.67	0.5	0.66

Information gain =

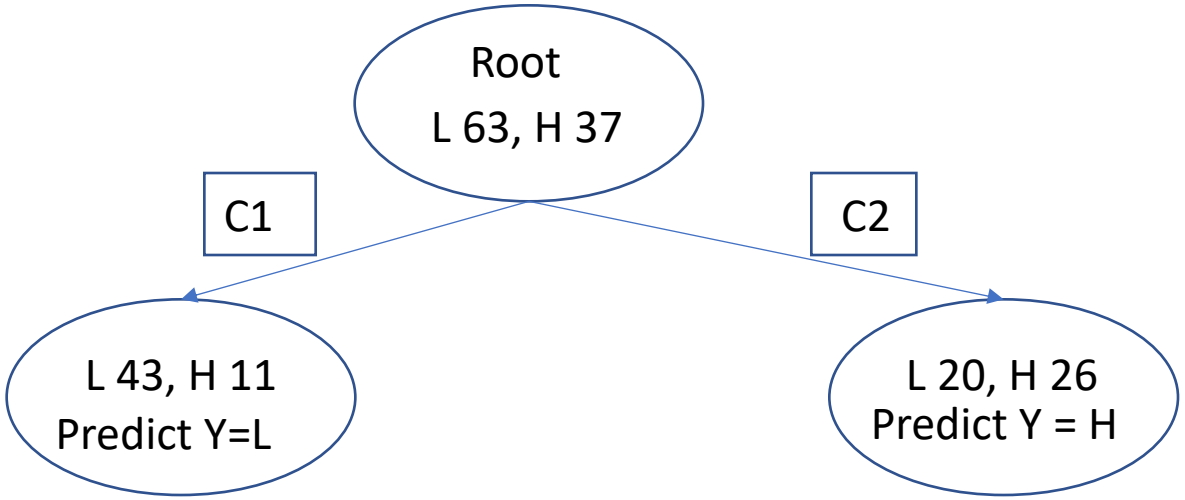
Original Entropy – (Split1_size*Split1_ entropy + Split2_size*Split2_ entropy)

0.66 – (90/100*0.67 + 10/100*0.5) ~ 0.01!!

Feature selection based on entropy

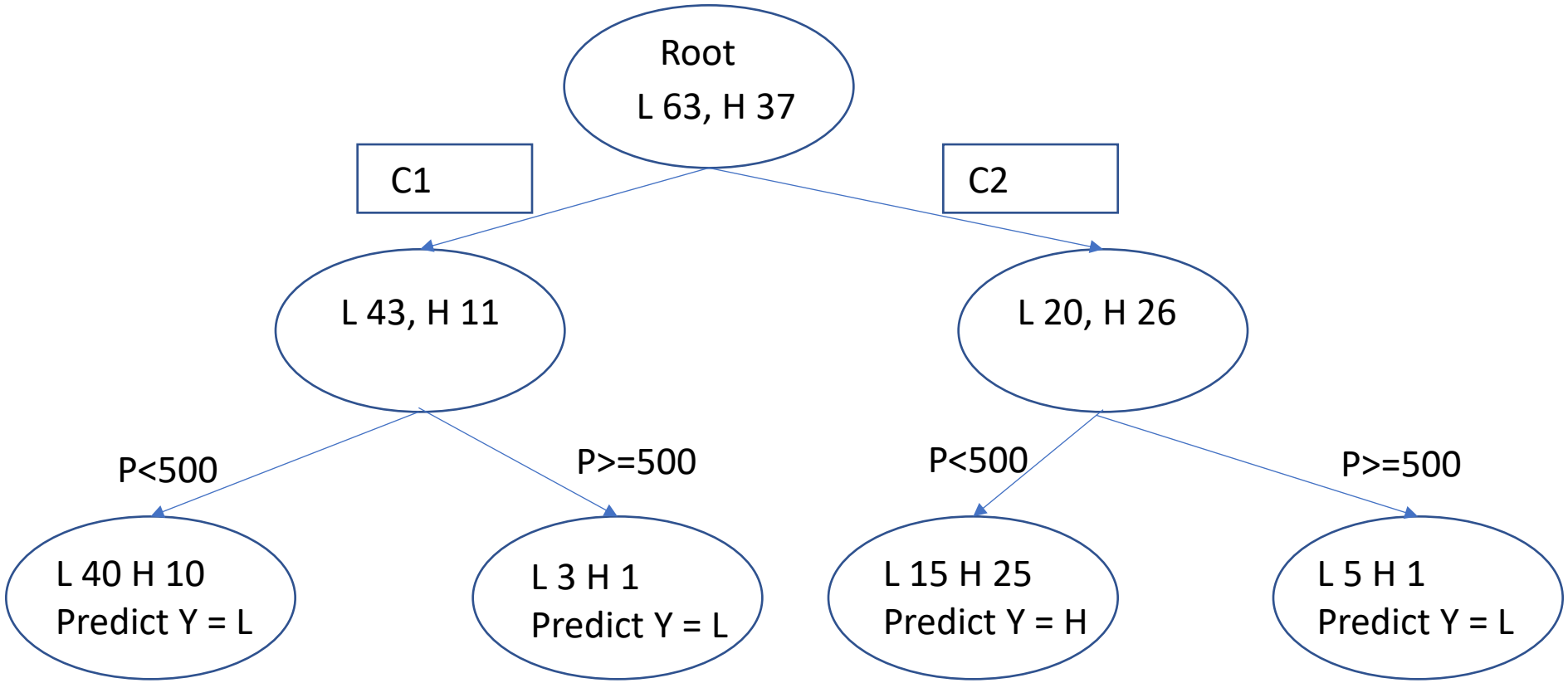
- Each discrete feature splits the dataset
- Continuous features can always be converted to discrete
- “Pure” dataset: - disbalanced class distribution
  - low entropy
  - high information gain
- Choose that feature which provides most information gain!

Decision Stump



Training accuracy: 43/63 for LOW, 26/37 for HIGH, 69/100 OVERALL

Decision Tree



- Does this split provide “information gain”???
- If yes, split. If no, stop at previous step

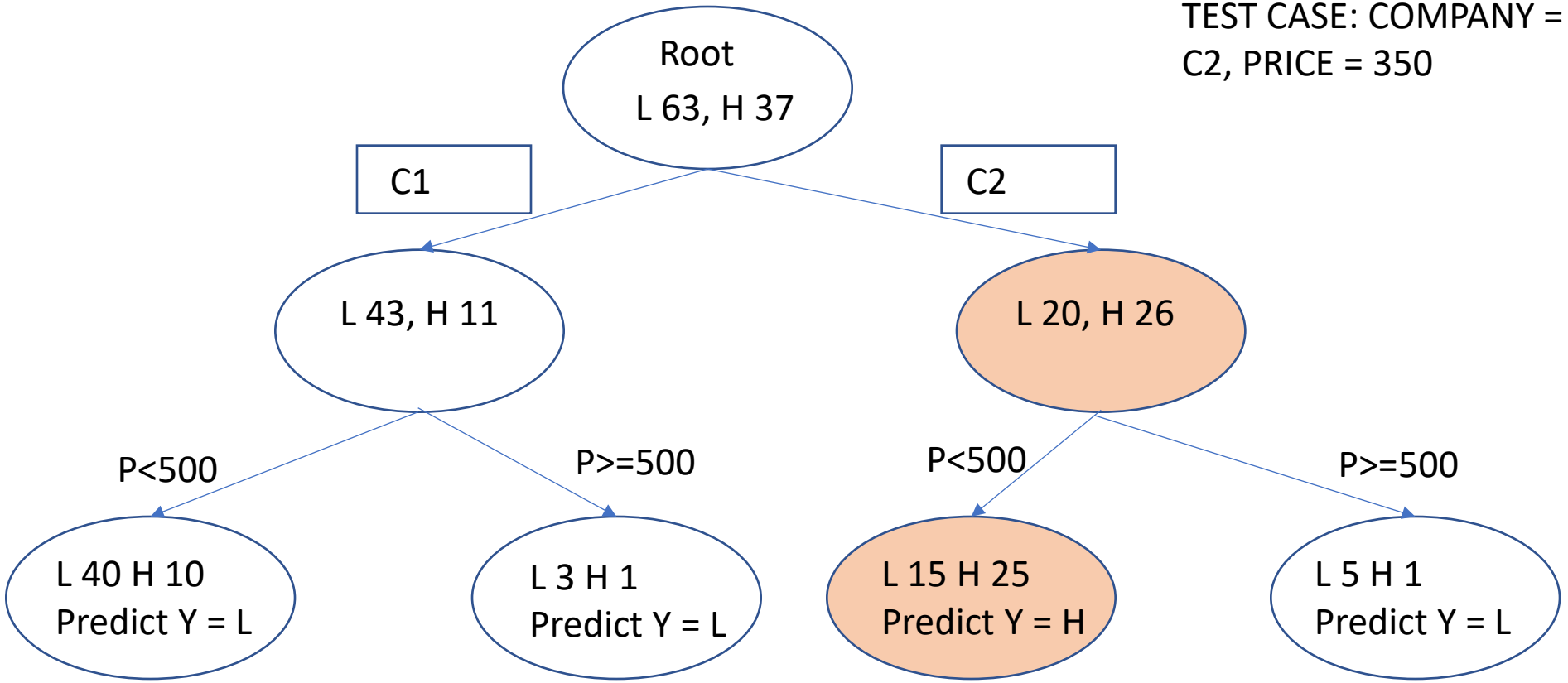
Decision Tree algorithm

- 1. Identify the feature that results in maximum information gain
- 2. Split the dataset accordingly
- 3. Identify if any feature can result in further information gain on the split sets
- 4. If yes, split further. If no, stop.
- 5. Goto 3
- 6. At each leaf, the prediction is the mode label

Test:

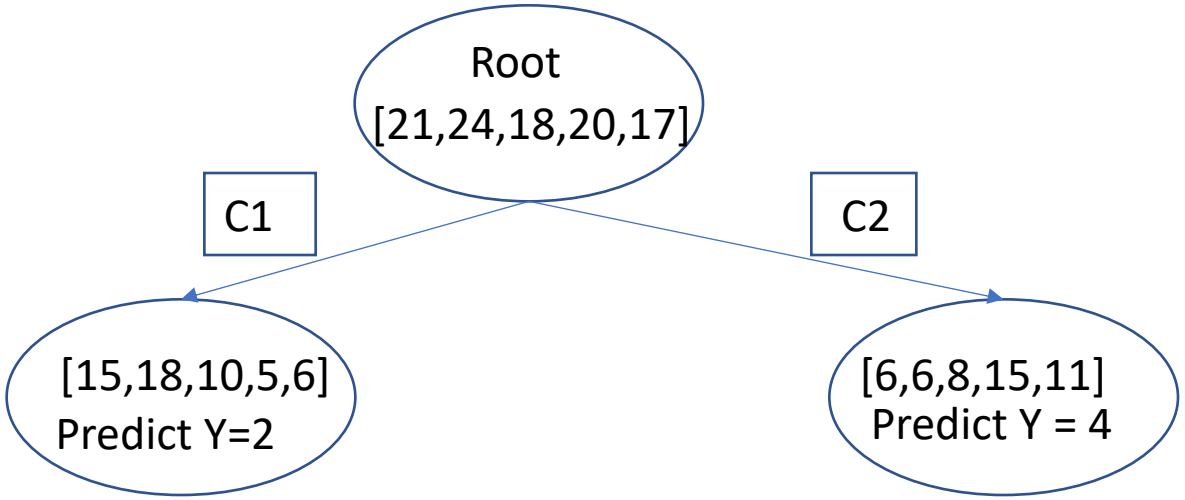
- Follow sequence of decisions based on features of test example
- Make prediction according to leaf

Decision Tree



- Prediction: Y= H

Multi-class Decision Stump



Training accuracy: 18/24 for CLASS 2, 15/20 for CLASS 4, 33/100 OVERALL

Advantages and Disadvantages

Advantage:

- Easy to interpret
- Easy to classify at test time
- Provides a ranking of features (according to usefulness)

Disadvantages:

- No optimal solution known, IG is just heuristic, can create many small branches
- Can cause overfitting if tree grows deep (need to stop growing)

Regression Trees

- What if we want to predict Average Rating of a product?
- Real number between 1 and 5!
- Decision trees can also be used for regression
- Measure of homogeneity at each node: variance of labels (instead of entropy)
- Split criteria: reduction in total variance (instead of information gain)
- Final prediction: Mean label in the leaf node (instead of mode)

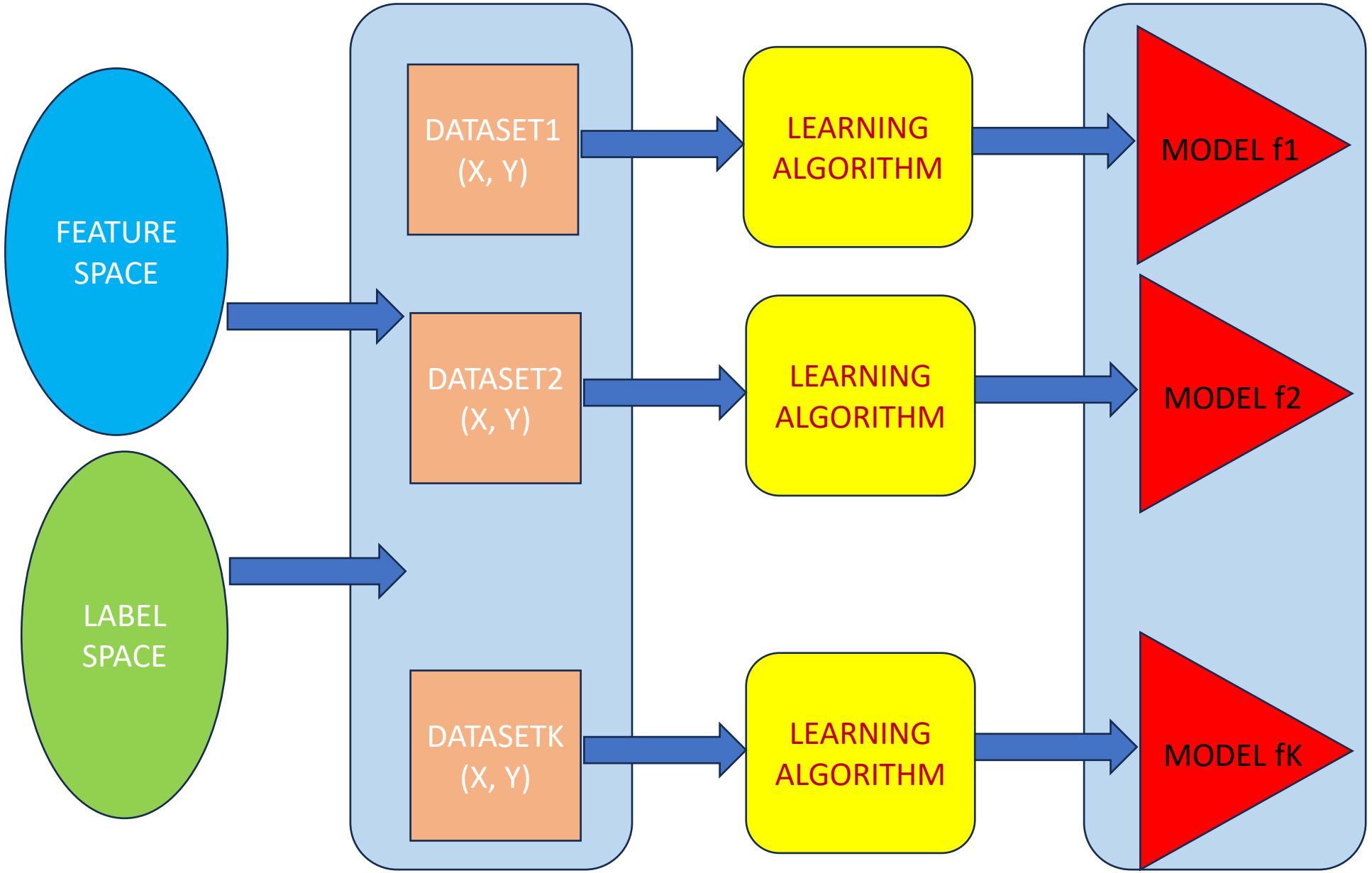
Variance Reduction for Regression

	COMPANY=C1	COMPANY=C2	NO SPLIT
COUNT	54	46	100
MEAN of RATINGS	3.0	4.0	3.46
VARIANCE of RATINGS	1.5	0.5	1.1
Reduction in Variance			1.1-(0.54*1.5+0.46*0.5)= 0.06

Variance Reduction for Regression

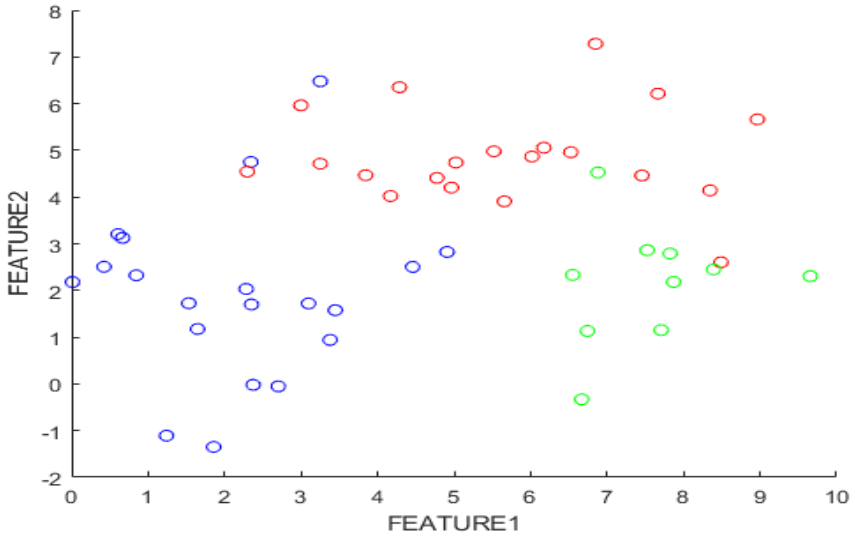
	VIDEO RATE <30 fps	Video RATE >=30fps	NO SPLIT
COUNT	70	30	100
MEAN of RATINGS	3.1	4.5	3.46
VARIANCE of RATINGS	1.2	0.4	1.1
Reduction in Variance			1.1-(0.7*1.2+0.3*0.4)= 0.14

Ensemble Supervised Learning



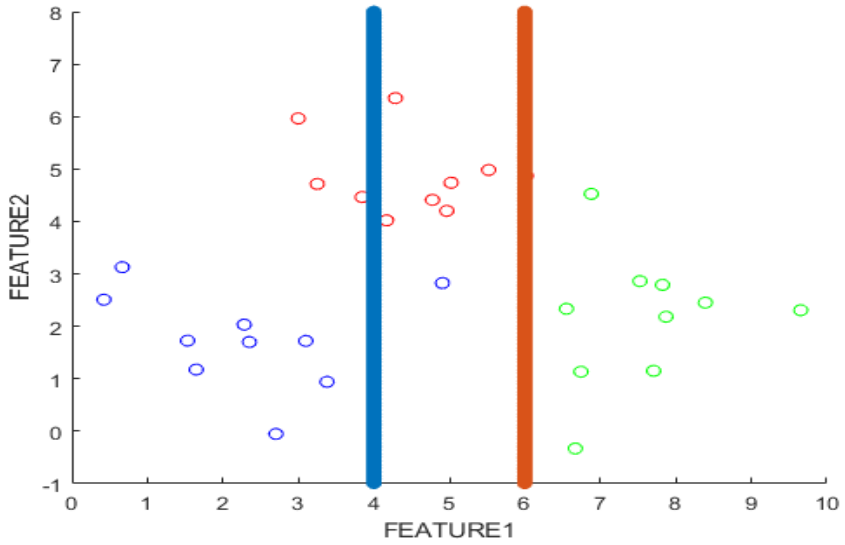
Multi-label classification

- A linear classifier creates 2 half-spaces
- So, it can do only binary classification!
- What for multi-label datasets?



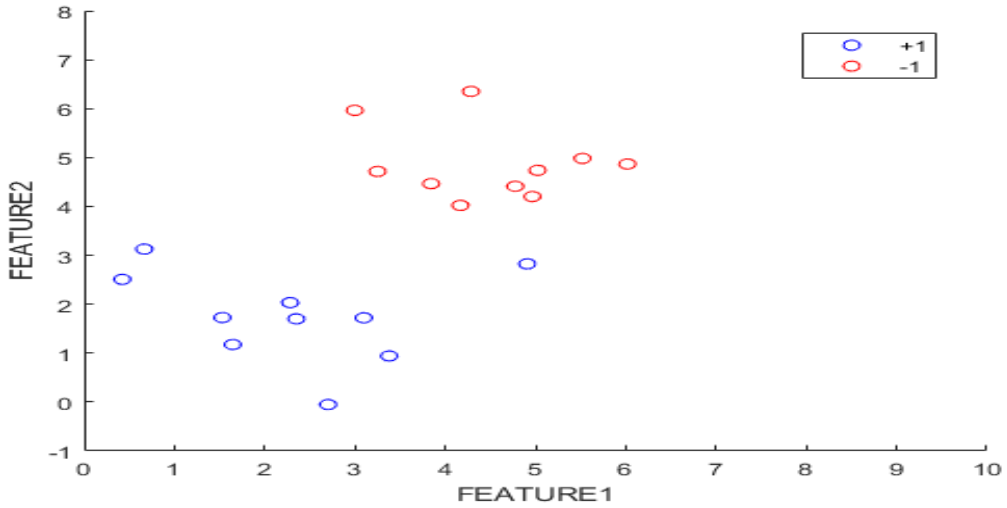
Multi-label classification

- A linear classifier creates 2 half-spaces
- So, it can do only binary classification!
- What for multi-label datasets?
- Multiple linear classifiers are needed!
- But how to find them?



One-vs-one

- Take any two classes, disregard the rest
- Consider any of them as +1, other as -1



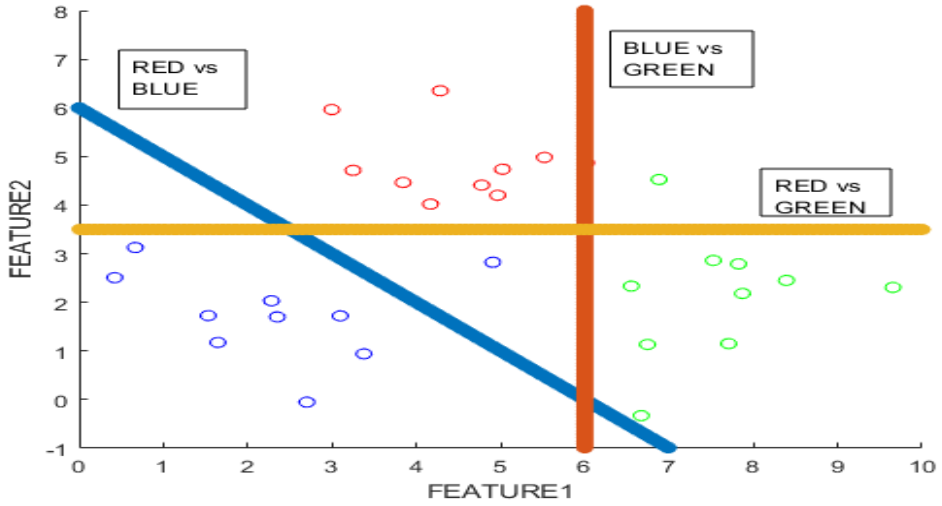
One-vs-one

- Take any two classes, disregard the rest
- Consider any of them as +1, other as -1
- Use perceptron algorithm to find

linear classifier 1!

- Repeat for other pairs of classes!
- Finally, get an “Ensemble” of

  (K-choose-2) classifiers!

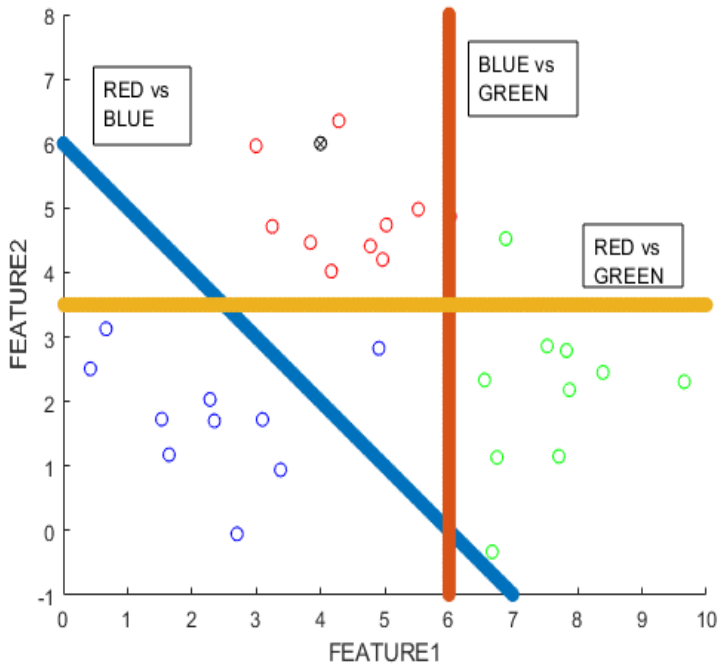


One-vs-one Classification

Given any new data-point, classify it with all one-vs-one classifiers!

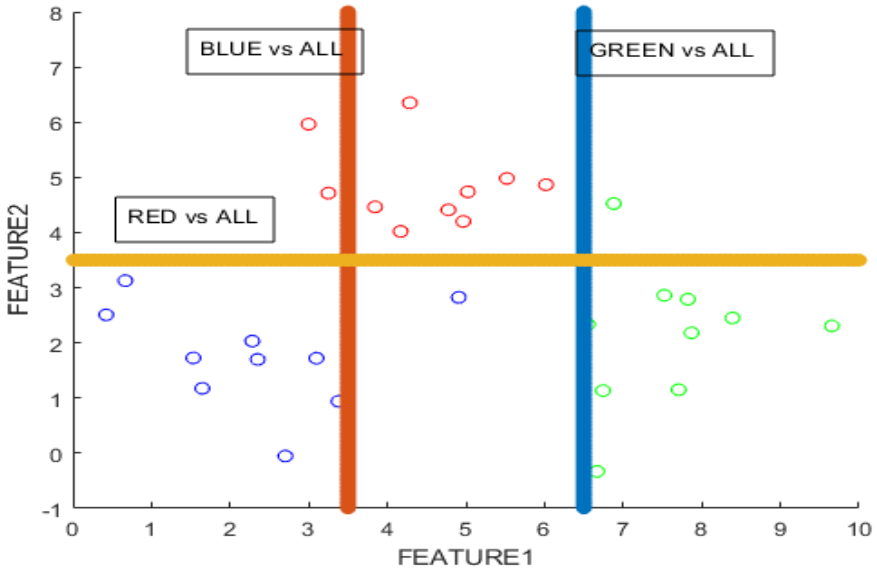
Take vote!

Classifier	Result	NET
RED vs BLUE	RED	RED-1, BLUE-0, GREEN-0
BLUE vs GREEN	BLUE	RED-1, BLUE-1, GREEN-0
RED vs GREEN	RED	RED-2, BLUE-1, GREEN-0
FINAL	RED!	



One-vs-all classification

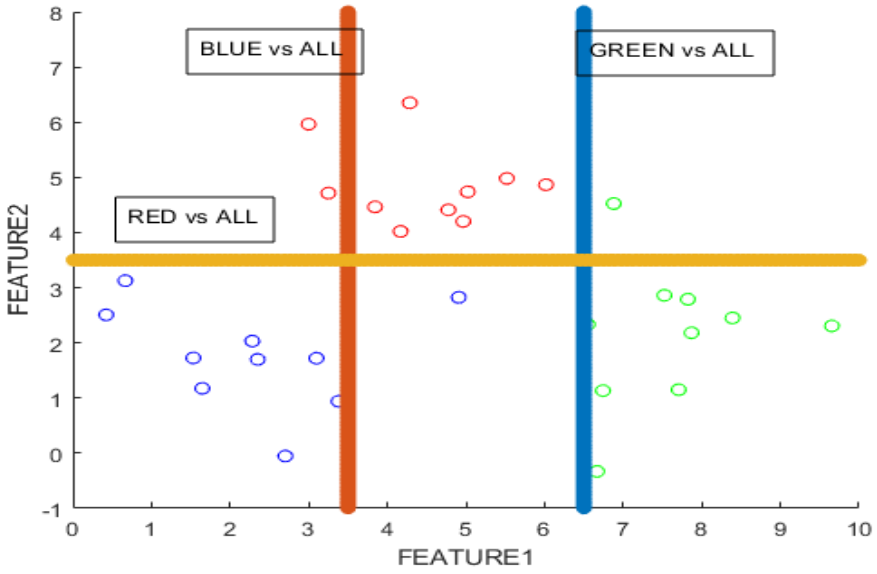
- Take one class, consider it +1
- All other classes: -1
- Find linear classifier by perceptron!
- Repeat for all classes, get ensemble



One-vs-all classification

- Take one class, consider it +1
- All other classes: -1
- Find linear classifier by perceptron!
- Repeat for all classes, get ensemble

Classifier	Result	NET
RED vs ALL	RED	RED-1, BLUE-0, GREEN-0
BLUE vs ALL	ALL	RED-1, BLUE-0, GREEN-0
GREEN vs ALL	ALL	RED-1, BLUE-0, GREEN-0
FINAL	RED!	



Different classifiers, different results

f1(X) = Y1

f2(X) = Y2

……

fK(X) = YK

- How to choose a single answer??

Simple Voting

f1(X) = Y1

f2(X) = Y2

……

fK(X) = YK

- How to choose a single answer??
- Classification: mode of all predictions!
- Regression: mean of all predictions!

Simple Voting

- How to choose a single answer??

classifier	f1	f2	f3	f4	f5	f6
Label	1	1	2	1	3	2

Label	Y=1	Y=2	Y=3
Total Votes	3	2	1

- Classification: mode of all predictions!
- Regression: mean of all predictions!
- Final prediction: Y =1 !

Simple Voting

- How to choose a single answer??

classifier	f1	f2	f3	f4	f5	f6
Label	1	1	2	1	3	2

Label	Y=1	Y=2	Y=3
Total Votes	3	2	1

- Classification: mode of all predictions!
- Regression: mean of all predictions!
- Final prediction: Y =1 !

Simple Voting

f1(X) = Y1

f2(X) = Y2

……

fK(X) = YK

- How to choose a single answer??

- Classification: mode of all predictions!
- Regression: mean of all predictions!
- But if all classifiers are not equally reliable????

Weighted Voting (Classification)

- Attach a “weight” with each classifier
- More reliable classifier -> higher weightage

classifier	f1	f2	f3	f4	f5	f6
Label	1	1	2	1	3	2
Weight	7	5	9	5	7	9

Weighted Voting (Classification)

- Attach a “weight” with each classifier
- More reliable classifier -> higher weightage

classifier	f1	f2	f3	f4	f5	f6
Label	1	1	2	1	3	2
Weight	7	5	9	5	7	9

Label	Y=1	Y=2	Y=3
Total Weight	17	18	7

- Final prediction: Y = 2!

Weighted Voting (Regression)

- Attach a “weight” with each classifier
- More reliable classifier -> higher weightage

classifier	f1	f2	f3	f4	f5	f6
Label	1.1	0.8	2.3	1.5	3.1	2.7
Weight	7	5	9	5	7	9

Weighted Voting (Regression)

- Attach a “weight” with each classifier
- More reliable classifier -> higher weightage

classifier	f1	f2	f3	f4	f5	f6
Label	1.1	0.8	2.3	1.5	3.1	2.7
Weight	7	5	9	5	7	9

- Weighted label = 1.1*7 + 0.8*5 + 2.3*9 + 1.5*5 + 3.1*7 + 2.7*6
- Total weight = 7 + 5 + 9 + 5 + 7 + 6

- Final prediction: Y = 2!

Weighted Voting (Regression)

- Attach a “weight” with each classifier
- More reliable classifier -> higher weightage

classifier	f1	f2	f3	f4	f5	f6
Label	1.1	0.8	2.3	1.5	3.1	2.7
Weight	7	5	9	5	7	9

But how to find these weights?

- assess performance of all models on a common validation set!

How to build ensemble of classifiers?

- Same training set, different classifier functions
- Split the dataset into multiple parts; train a classifier on each part
- Select different subsets of features; train a classifier for each subset
- Bootstrap Aggregation (BAGG-ing)!
- Final prediction from ensemble mean or mode -> reduces variance!

How to build ensemble of classifiers?

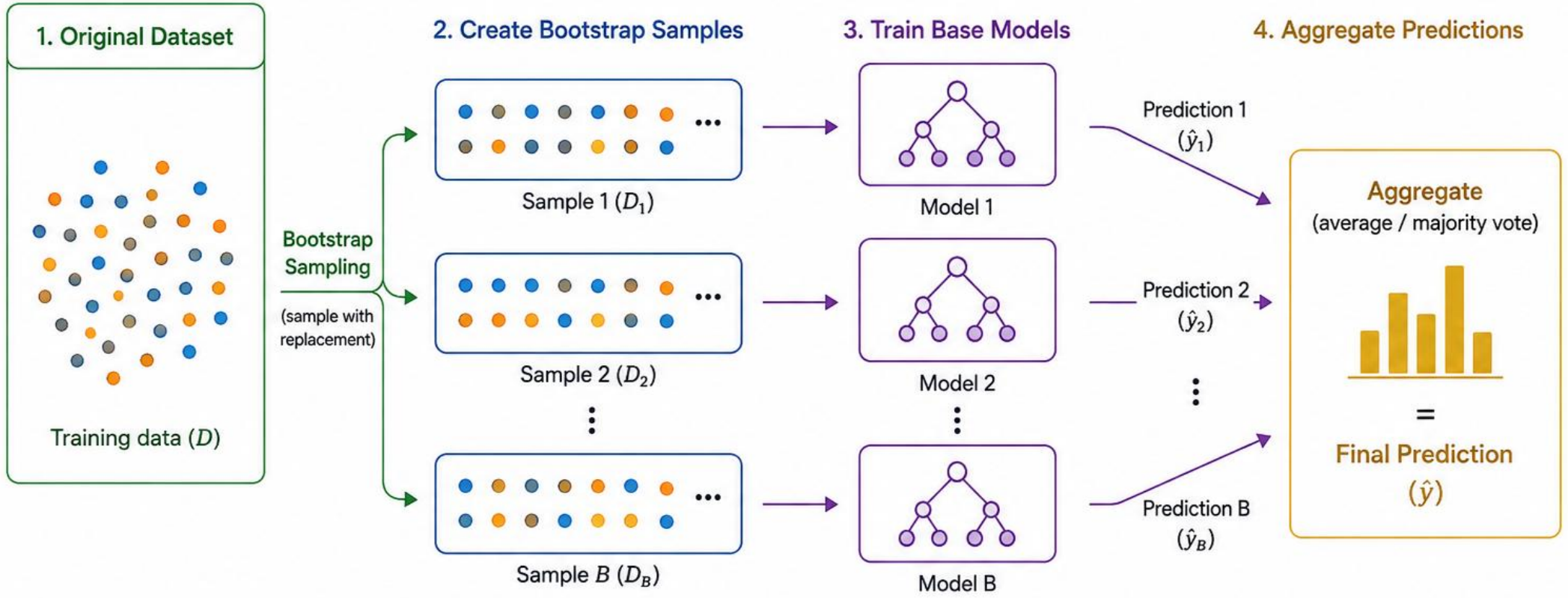
- Same training set, different classifier functions
- Split the dataset into multiple parts; train a classifier on each part
- Select different subsets of features; train a classifier for each subset
- Bootstrap Aggregation (BAGG-ing)!
- Final prediction from ensemble mean or mode -> reduces variance!

Var((X1 + X2)/2) = ¼*Var(X1) + ¼*Var(X2) + ½*Cov(X1,X2)
Var((X1+X2)/2) < Var(X1), Var(X2) [Assuming Cov(X1, X2) = 0]
Extending, Var((X1+…..+XK)/K) < Var(X1), …… Var(XK)

i.e. Mean Random Variable has less variance than original Random Variables

Bootstrap Aggregation

- Use BAGG to generate K “versions” of original training set
- Train a classifier on each version
- Give weightage to each classifier according to its accuracy on validation set



Bootstrap Aggregation

- Given a training set of size N:
- Choose N examples from it with replacement
- i-th draw: each training example may be chosen with probability 1/N
- Same example may be chosen multiple times!

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X6	X3	X4	X10	X3	X7	X3	X6	X8	X2
X4	X7	X9	X5	X3	X6	X2	X6	X1	X6

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- Prob(X_j is never chosen) ~ 0.36
- Nearly one-third of original dataset will not be present in new dataset!

BAGGing of features

- Instead of choosing subsets of training examples
- We can choose subsets of the features!

- Each “version” has all the training examples, but only some of the features!

X(1)
X(2)
X(3)
X(4)
X(5)

X(4)
X(2)
X(3)
X(3)
X(2)

X(2)
X(1)
X(4)
X(3)
X(3)

X(4)
X(2)
X(3)
X(2)
X(3)

Random Forest

- Instead of choosing subsets of training examples
- We can choose subsets of the features!

- Each “version” has all the training examples, but only some of the features!

- If the classifier used is Decision Tree, then this approach is called Random Forest
- RF is one of the more powerful classifiers around

Random Forest

